

Abdoali Mazhar Zakir

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PROFESSIONAL SUMMARY

Final-year Software Engineering (Honours) student at UNSW with experience in full-stack development, cloud systems, and data-driven applications. Worked on a financial crime knowledge-graph project, turning large transactional datasets into structured, queryable insights. Strong grasp of the SDLC, including API development, automated testing, CI/CD, and Agile teamwork.

EDUCATION

Bachelor of Engineering (Honours) in Software Engineering
University of New South Wales

September 2024 – December 2026

Bachelor of Science in Computer Science & Engineering
International Centre for Applied Sciences (ICAS),

July 2022 – May 2024

RELEVANT EXPERIENCE

Project Engineer Intern

Tata Consultancy Services (TCS)

January 2026 – February 2026

- Built a Financial Crime Compliance knowledge graph by transforming large-scale transactions and account datasets into structured graph entities using Neo4j, enabling advanced network analysis for fraud detection.
- Applied graph similarity algorithms to identify hidden relationships between accounts, supporting fraud-risk analysis across complex financial networks.
- Collaborated within a multi-disciplinary engineering team following Agile delivery practices, contributing to project planning, milestone reviews, and final technical presentations to industry stakeholders.

Software Development

Enigma Technical Society

September 2023 – May 2024

- Engineered internal web tools and automation scripts using Python, Node.js, and Git, streamlining society operations and reducing manual workflows by 30%.
- Collaborated in a 6-member Agile team to design and maintain backend systems with Express.js and PostgreSQL, ensuring scalable data management and smoother event coordination.
- Deployed continuous integration pipelines using GitHub Actions, improving version control efficiency and reducing integration errors.
- Contributed to brainstorming sessions and technical documentation, reinforcing code quality and collaborative development practices.

PROJECTS

TerraCast – Viticulture Weather & Advisory Platform
UNSW

February 2026 – May 2026

- Designed and integrated serverless data pipelines using AWS Amplify to process and retrieve weather data (rain, frost, temperature, wind), supporting forecast ingestion and risk-based insights.
- Developed responsive frontend features using Next.js (App Router), including dashboards and recommendation views, with state management and API integration.
- Implemented server-side generative AI features using OpenAI API (gpt-4o-mini), including a viticulture assistant and forecast-based recommendation system using structured outputs.
- Collaborated in an Agile team, contributing to API design, system integration, and technical documentation, including testing strategy and deployment planning.

TOOHAK – Gamified Quiz Application
UNSW

February 2025 – May 2025

- Developed a Kahoot-inspired gamified quiz platform to improve student engagement and interactivity.
- Built real-time multiplayer features and session management using Express.js APIs integrated with a React front-end.
- Conducted API testing with Postman and unit testing using Jest, achieving full backend coverage and error-free integration.
- Collaborated in a team-based Agile workflow, performing sprint planning and continuous feedback cycles.

UNSW Cloud-Deployed Invoice Management Backend API

February 2025 – May 2025

- Designed, developed, tested, and deployed backend APIs following SDLC practices.
- Collaborated with a cross-functional team to design and implement RESTful APIs using Node.js and Express, supporting reliable backend services.
- Integrated PostgreSQL database with validation and authentication middleware to ensure secure data handling.
- Containerised the application using Docker and deployed it to AWS, achieving reliable CI/CD testing environments.
- Performed API testing using Postman and conducted peer code reviews to maintain high code quality.

Thesis — Citation Prediction using CDCHGN Knowledge Graphs

2026

- Designing heterogeneous graph neural network models to predict citation links using community detection and node embedding techniques — large-scale graph data processing directly applicable to MongoDB's Atlas Graph and aggregation pipeline internals.

UNSW Automated Testing & Fuzzing Framework

June 2025 – August 2025

- Implemented a Python-based testing framework to automate integration tests for a simulated airline check-in system.
- Designed fuzz testing modules generating randomised edge cases, uncovering 12+ input validation vulnerabilities and improving system robustness.
- Increased overall test coverage by 45%, reducing manual QA effort and improving code reliability.

CORE COMPETENCIES

Languages: Python, JavaScript, C++, SQL, Cypher, HTML/CSS.

Frameworks & Libraries: Node.js, React, Jest, Pytest, Pytorch

Databases: PostgreSQL, MySQL

Tools & Platforms: Docker, AWS (TerraForm, S3), Postman, Neo4j

VOLUNTEERING AND EXTRACURRICULAR

Lead Volunteer
Voluntary Service Overseas (VSO), Manipal

June 2023- May 2024

Secretary
Social Football Society, UNSW

March 2024- Present

Sports Captain
International Centre for Applied Science, Manipal University

June 2023- May 2024